AS2111 Stochastic Models
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Group Coursework

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Group Z

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1. Introduction

This report examines an insurance company's financial stability by modelling the evolution of its fund over 40 quarters. The fund changes each quarter due to random income, investment returns and claim losses. Using normal approximations and Monte Carlo simulation, we estimate ruin probabilities under varying assumptions, including business line dependence and differing economic conditions.

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2. Model setup

The insurer's fund evolves according to:

$$\boldsymbol{F_n = F_{n-1} + A_n - L_n, \qquad n = 1, \dots, 40}$$

where:

- A_n= quarterly asset income
- L_n= total quarterly loss
- A_n and L_n are independent & i.i.d.

All amounts are expressed in millions.

2.1 Liabilities

Quarterly losses come from three independent business lines:

$$\boldsymbol{L_n = L_{n,1} + L_{n,2} + L_{n,3}}$$

Each $L_{n.i}$ is lognormally distributed with the following means and variances:

Business Line	Mean (μ_i)	Variance(σ_i^2)
1	10	4
2	15	9
3	8	2

2.2 Assets

The assets (A_n) follow a gamma distribution with:

$$\boldsymbol{E[A_n] = E[L_n] \cdot (1 + \lambda)}$$

$$\boldsymbol{VAR[A_n] = VAR[L_n] \cdot \kappa}$$

$\lambda = 4\%$ (percentage loading),
$\kappa = 75\%$ (risk factor for asset variance)

2.3 Initial Capital and Quantities of Interest

The initial fund is:

$$\boldsymbol{F_0 = \alpha\, E[A_n], \qquad \alpha = 0.5,}$$

where $\alpha = 0.5$ is the proportion of expected quarterly premium income used as initial capital.
The two ruin probabilities of interest are:

$$\boldsymbol{\pi 1 = P(F_{40} < 0), \qquad \pi 2 = P(F_{4n} < 0) for\ some\ n = 1, \dots, 10}$$

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3. Use the normal approximation for a random walk to calculate π_1 (part a)

Step 1: Total Liability and asset moments

From the business line parameters in the model setup, total quarterly loss has

$$E[L_n] = 33\,, \quad VAR[L_n] = 15$$

Using the asset relations in **2.2** this gives

$$E[A_n] = 34.32, \qquad VAR[A_n] = 11.25$$

Step 2: One quarter increment of the fund

We define the quarterly increment of the fund as

$$Y_n = A_n - L_n.$$

Since A_n and L_n are independent & i.i.d:

$$E[Y_n] = E[A_n] - E[L_n] = 1.32, \qquad VAR[Y_n] = VAR[A_n] + VAR[L_n] = 26.25$$

Step 3: Mean and Variance of F_{40}
After 40 quarters can be written as:

$$F_{40} = F_0 + \sum_{n=1}^{40} Y_n, \qquad F_0 = 0.5\, E[A_n] = 17.16$$

Using the properties of the sums of i.i.d variables,

$$E[F_{40}] = F_0 + 40E[Y_n] = 69.96, \qquad VAR[F_{40}] = 40VAR[Y_n] = 1050$$

Step 4: Normal approximation and ruin probability
By the Central Limit Theorem, we can approximate

$$F_{40} \approx N(69.96, 1050).$$

The ruin probability at the 10- year horizon is therefore:

$$\pi_1 = P(F_{40} < 0) = \Phi\left(\frac{0 - 69.96}{\sqrt{1050}}\right) = 0.0154_{\,4d.p}$$

Thus, under the normal approximation, the probability that the fund is negative after 40 quarters is around **1.54%.**

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4. Monte Carlo Method to estimate π_1 and π_2 (part b)

This section describes the application of the Monte Carlo simulation method to estimate the two ruin probabilities of interest noted in section **2.3**. Since the Monte Carlo is an estimate of probability based on counting the proportion of simulations in which ruin occurs, its accuracy depends on the number of simulations. To make sure the estimate was reliable, a target margin of error of 0.1% at the 95% confidence interval was used. To calculate the number of simulations required, the absolute margin of error approach was used as it was easy to interpret, remained consistent across estimate and tied directly to the sample size.

4.1 Converting losses to lognormal parameters (Each business line is modelled as lognormal):
The means and variances of the losses are given(μ, σ^2), so we need to convert them to lognormal parameters $(\mu_{log}, \sigma_{log})$ using:

$$\sigma_{log}^2 = \log\left(1 + \frac{VAR(L)}{E[L]^2}\right), \quad \mu_{log} = \log(E[L]) - \frac{1}{2}\sigma_{log}^2$$

4.2 Compute the parameters for the Gamma distribution of assets:
Gamma (shape, scale)

$$shape = \frac{\mu_A^2}{Var(A)}, \qquad scale = \frac{Var(A)}{\mu_A}$$

4.3 Formula for the absolute margin of error for a proportion is defined as:

$$E = Z_{\propto/2}\sqrt{\frac{\hat{\pi}(1-\hat{\pi})}{M}}$$

This is then solved for M to get the number of simulations using the accuracy mentioned above to get M = 58,340.

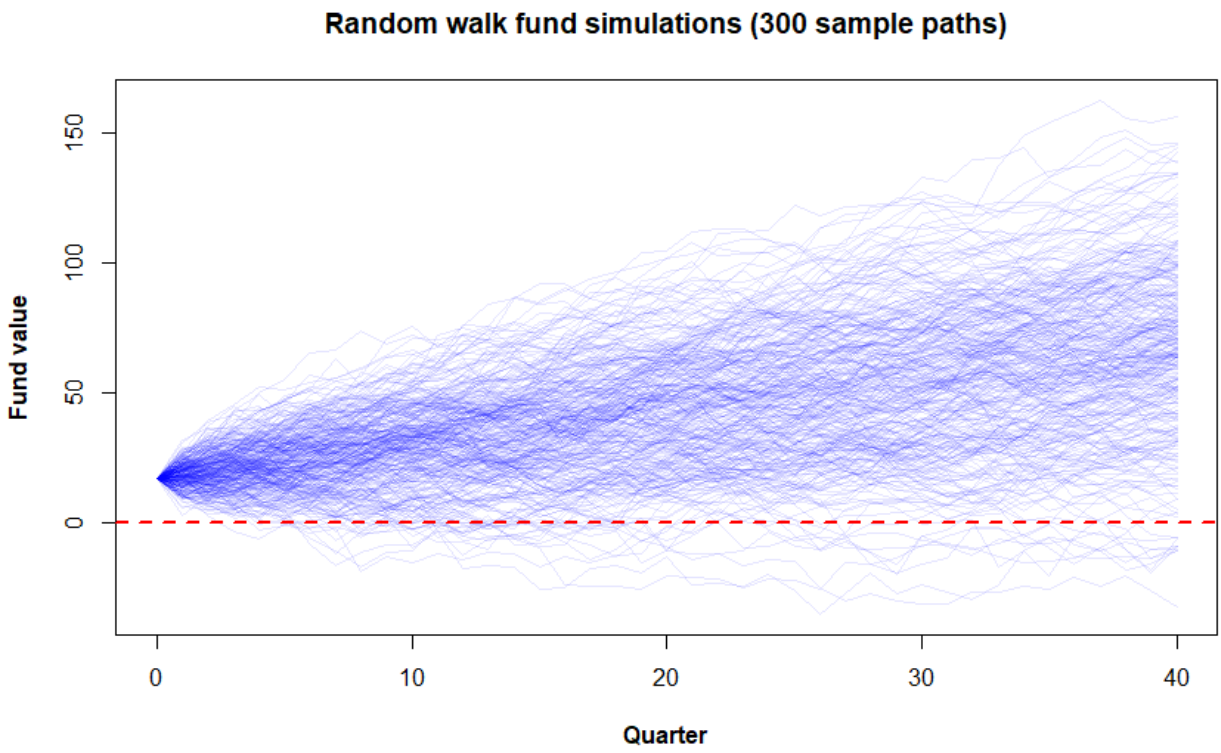
4.4 The fund is then simulated

4.5 Estimation of the ruin probabilities: $\pi_1 = 0.01589, \ \pi_2 = 0.09210$

In part A π_1 was calculated using the normal approximation and returned a value of 0.0154 which is slightly lower than the value calculated above using the Monte Carlo simulation. This is because the normal approximation in Part A assumes symmetric, light-tailed behaviour while the Monte Carlo simulation uses the true heavy-tailed lognormal losses which allow for rare, very large losses. The year-end ruin (π_2) is much higher, reflecting the fact that ruin can occur early in the horizon even if the final fund value is positive.

4.6 Plot of the sample paths

300 random sample paths are plotted to illustrate the stochastic behaviour of the fund, with a red horizontal line marking ruin at 0:



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5. Ruin Probability Under Correlated Losses (part c)

This section investigates how the ruin probability π_1 changes if the losses of business lines 1 and 2 are correlated. For each correlation value, the Monte Carlo method (using n=58,384, as determined in part b) was used to estimate ruin probability π_1.

5.1 Correlated lognormal framework

The end of quarter losses are now specified as:

$$\log L_{n,1} = \mu_1 + \sigma_1 Z_{n,1}\,, \qquad log L_{n,2} = \mu_2 + \sigma_2\,(\rho Z_{n,1} + \sqrt{1-\rho^2}\,Z_{n,2})$$

$Z_{n,1}$ and $Z_{n,2}$ are underlying standard normal drivers.

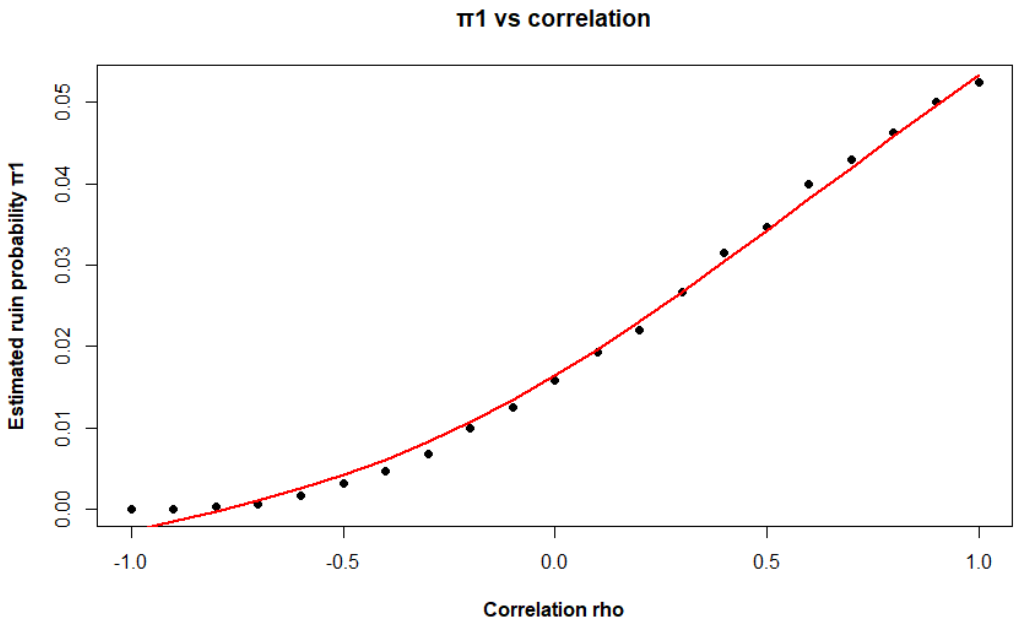
5.2 Asset Parameters Distributions

A crucial step of the process was to update the parameters of the asset distribution for each ρ as the Gamma parameters depend on the mean, $E[A_n]$, and variance, $VAR[A_n]$, which in turn depended on the mean, $E[L_n]$,and variance, $VAR[L_n]$, of the total loss. The correlation ρ only impacts the variance of total loss. We used the following equation:

$$VAR[L_n] = \sum_{i=1}^{3} VAR[L_i] + 2Cov(L_1, L_2)$$

5.3 Loop over correlation values using the ρ-dependent parameters. The script evaluates 21 correlation values and produced the following:

ρ	π_1
-1.0	0.0000171409
-0.9	0.0000342818
-0.8	0.0002056908
-0.7	0.0005827905
-0.6	0.0016798080
-0.5	0.0031196435
-0.4	0.0047137470
-0.3	0.0067020912
-0.2	0.0098560165
-0.1	0.0124614330
0.0	0.0157696263
0.1	0.0193006514
0.2	0.0220089133
0.3	0.0266026740
0.4	0.0314364073
0.5	0.0346246143
0.6	0.0399897155
0.7	0.0429208091
0.8	0.0462290024
0.9	0.0500685636
1.0	0.0524511484



Using the values in the table, we plotted a graph with a line of best fit. The plot clearly showed that the ruin probability π_1 is a monotonically increasing function of the correlation coefficient (ρ).

5.4 Impact of Loss Correlation (ρ) on ruin probability ($\pi_{1)}$

The overall find is that the ruin probability goes up as the correlation goes up.

- **When Losses are Synchronized (Positive Correlation):** The simultaneous occurrence of major losses drastically increases the overall unpredictability (variance) of the total claims paid out. Because the insurer's assets are tied to this volatile liability, the fund itself becomes more volatile, increasing the chance of it running out of money.

- **When Losses Offset (Negative Correlation):** By having losses that rarely occur at the same time, the insurer is effectively spreading the risk. This stable flow of money minimises the total volatility of claims paid, making the fund much safer and reducing the capital needed to remain solvent.

6. Ergodicity and stationary distribution (part d(i))

A finite Markov chain is ergodic if it is both aperiodic and irreducible.

Transition Matrix

	N	G	B	T
N	0.75	0.15	0.05	0.05
G	0.90	0.10	0.00	0.00
B	0.50	0.00	0.45	0.05
T	0.50	0.00	0.50	0.00

Irreducibility:

From the transition matrix, each of the states G, B and T can move to N in one step, and from N there is a positive probability of moving to all other states. Hence all states communicate and the chain is irreducible.

Aperiodicity:

State N has a self-transition probability $p_{NN} = 0.75$, so its period is 1. In an irreducible Markov chain, all states share the same period, so the entire chain is aperiodic.
Since the chain is both irreducible and aperiodic it is ergodic.

Stationary Distribution:

Since the chain is ergodic, it has a unique stationary distribution

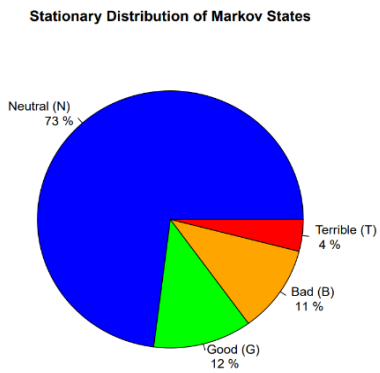
$$\pi = (\pi_N, \pi_G, \pi_B, \pi_T)$$

satisfying the balanced equation:

$$\pi = \pi P, \qquad \pi_N + \pi_G + \pi_B + \pi_T = 1.$$

Solving the system in the transition matrix gives the stationary distribution:

$$\pi \approx (0.732, 0.122, 0.105, 0.042).$$



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7. Ruin Probability under Markov-Dependent Economic Conditions (part d(ii))

To estimate the ruin probability under the Markov-chain economy, we use the same Monte-Carlo simulation method as in part b. The modelling of losses, the random walk update and the definition of ruin, fund negative at time 40, all remain unchanged. The key difference is that the asset parameters mentioned in **2.2** (λ and κ) now depend on the economic state of each quarter.

We assume four economic states **N**(Neutral), **G**(Good), **B**(Bad) and **T**(Terrible) with the following parameters:

$$\lambda = (0.04, 0.06, 0.02, 0.02), \qquad \kappa = (0.75, 0.60, 0.75, 1.00), \qquad states\ ordered\ as\ (N, G, B, T)$$

We start in state N, and each next state is generated using the transition matrix in part d(i) so the sequence of 40 quarterly states follows the Markov chain.

Thus, for quarter n in state $s \in \{\boldsymbol{N}, \boldsymbol{G}, \boldsymbol{B}, \boldsymbol{T}\}$, the corresponding values of λ_s and κ_s are obtained. These then determine the mean and the variance of the Gamma- distributed asset income:

$$E[A_n] = E[L_n] \cdot (1 + \lambda_{S_n}), \qquad VAR[A_n] = \kappa_{S_n} \cdot VAR[L_n]$$

with $E[L_n]$ and $VAR[L_n]$ calculated earlier.

The quarterly losses are simulated in the same manner as in part (b), using the sum of the three independent lognormal business line losses defined in section **2.1**. The simulation begins from the initial fund level in section **2.3**, and the fund evolves quarterly according to the random-walk equation given in section **2.0**. For each simulated path, we record whether the final fund value is negative. Repeating this over many simulations gives the Monte Carlo estimate of the ruin probability:

$$\hat{\pi}_1^{(D)} = \frac{Number\ of\ ruined\ paths}{total\ number\ of\ paths}$$

Results

The Monte Carlo estimate of the ruin probability under the Markov-state model is:

$\hat{\pi}_1^{(D)} = 0.01671,$ $\quad Standard\ Error = 0.00053,$ $\quad$ 95% CI $(0.01567,\ 0.01775)$.

The ruin probability under the Markov model, $\hat{\pi}_1^{(D)} = 0.01671$, is higher than the Monte Carlo baseline $\pi 1$ value, 0.01589, giving an increase of around 0.00082 percentage points. Although the Markov chain spends most of the time in the Neutral State ≈ 73%, the stationary distribution also assigns meaningful probabilities to the Bad and Terrible states.

These adverse states reduce the expected asset income through lower λ values, and in the Terrible state they also increase volatility through higher κ values. Consequently, during these periods the insurer's fund grows more slowly and becomes more vulnerable to large losses. Since the chain is ergodic the model is guaranteed to enter bad states within the 40-quarter horizon, when these adverse states occur consecutively, they create runs of weak performance that make it more likely for the fund level to drop and eventually fall below 0. This explains why the ruin probability for the Markov dependent model is higher than the i.i.d baseline figure.